

A crop makes a pest-control protein

Learning intention

Describe genetic engineering and use evidence to weigh a specific benefit, a risk and an unanswered question.

What changed: the DNA, the whole species, or the weather?

Today: explain the change and assess a trial.

Retrieve genes and selection

R1. What is a gene?

R2. What do humans choose in selective breeding?

R3. Can a resistance trait become more common over generations?

What genetic engineering means

Genetic engineering: alter the genome

A genome is all an organism's genetic information.

**Add a
segment**



Change DNA



Delete DNA

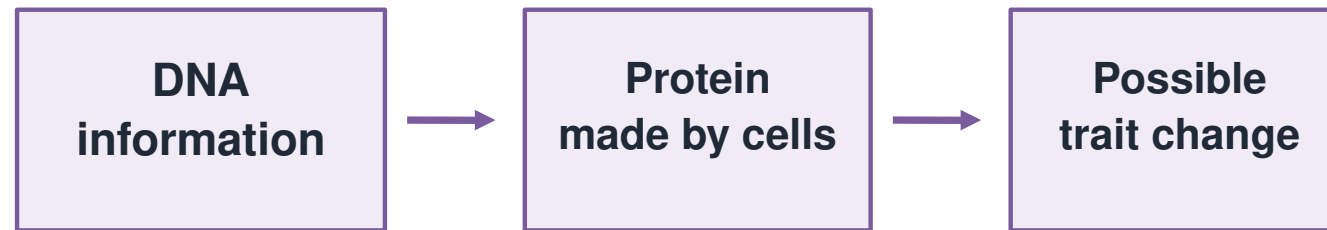


Cross-species transfer is only one method.

Laboratory technologies alter the genome directly. DNA may be added, changed or removed. A gene transfer is one example, not the only method.

Connect the change to the trait

Connect a genetic change to its effect



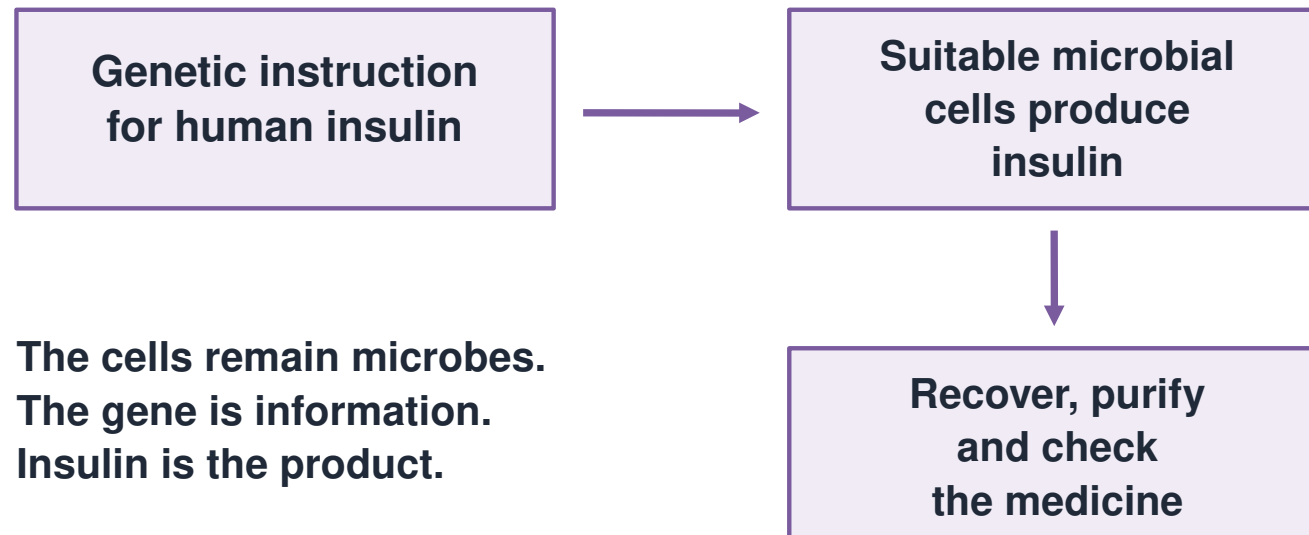
A Bt gene can enable a crop to make a protein that harms certain insect pests.

Simplified model; the change must be checked.

A gene can affect the protein a cell makes. A Bt crop can make a protein harmful to certain insect pests. Testing checks whether the intended effect occurs.

The product is the protein

Example: make a useful medicine



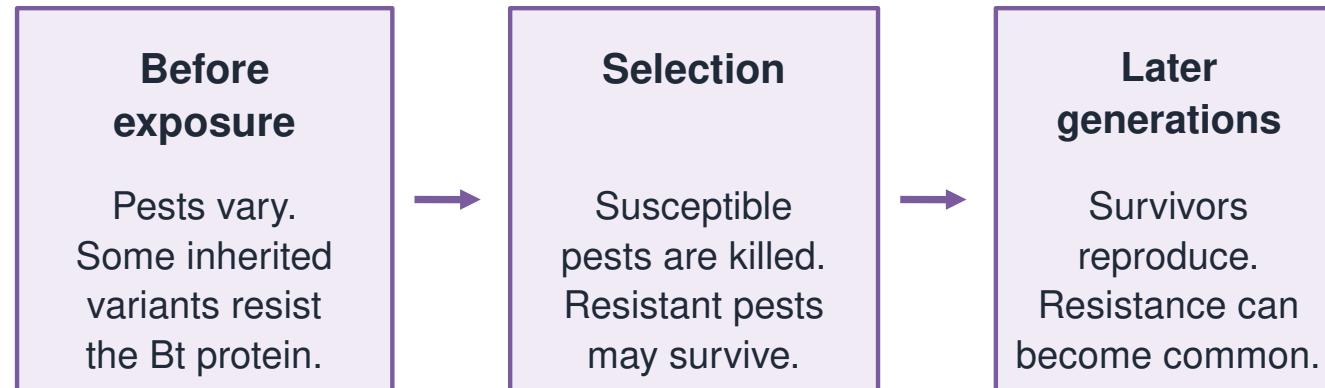
**The cells remain microbes.
The gene is information.
Insulin is the product.**

Conceptual model, not a practical method.

Engineered microbial cells can make human insulin. The product is recovered, purified and checked. The cells do not become human.

Predict a risk using biology

Why resistance needs monitoring



Monitoring and resistance-management strategies can delay resistance, but not promise zero risk.

Pest populations contain inherited variation. Resistant survivors can reproduce. Resistance may spread, so monitor it.

Trial board: what does the evidence show?

W1. Compare damage and yield in this trial. W2. Which claim is justified: promise, success here, or proof everywhere? Use exact quantities.

Crop	Target-pest damaged plants /100	Mean grain yield (t/ha)
Comparison non-GM	30	6.9
Bt GM	8	7.9

Constructed sample data: means from three matched plot pairs at one site in one season; neither crop received target-insect spray. Other growing conditions matched.

Spend your two next checks

W3. Choose two: repeat sites/seasons; monitor resistance; measure crop colour; count likes online.

Defend how each choice reduces uncertainty.

Acknowledge a further question your choices leave open.

Which explanation is scientifically sound?

Choose the statement you would keep in a textbook.

- A** Engineering always transfers DNA between species.
- B** An insulin-producing bacterium becomes human.
- C** Engineering alters DNA and may introduce a useful characteristic.
- D** Engineering guarantees that all offspring are safer.

Write a provisional recommendation

Use the mechanism and crop-trial evidence.

Recommend a next step with a benefit, risk and missing check.

Choose your response route with the teacher.

A recommendation is more than a vote

In this trial, ... so a possible benefit is ...

However, ... could happen because ...

I recommend ... while checking ...

Optional Higher-tier extension

Ready for more? Use the extra mechanism box on the Stretch page.

Explain a vector, restriction enzyme and ligase.

This detail is not needed for today's core success.

Review a recommendation fairly

Does it explain the genetic change?

Does it separate evidence from uncertainty?

Use one audience question to sharpen the next step.

Carry a good question into Week 4

Which crop and trait are being assessed?

What evidence would answer our unanswered question?

Who would be affected by the decision?

Explain, support, qualify

- E1. Define genetic engineering in one sentence.
- E2. Give one measured benefit in the trial.
- E3. Give one risk and a useful next check.